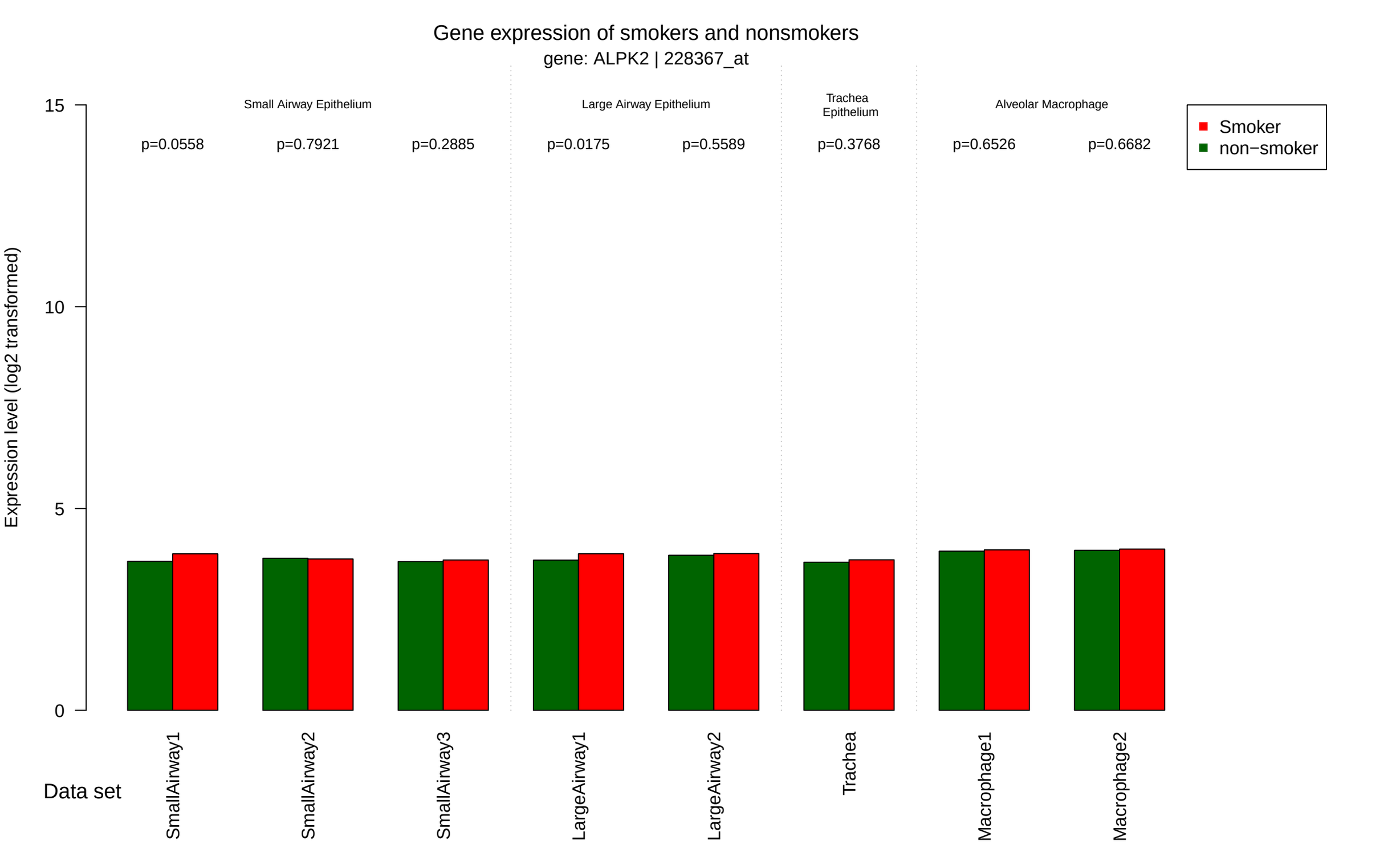
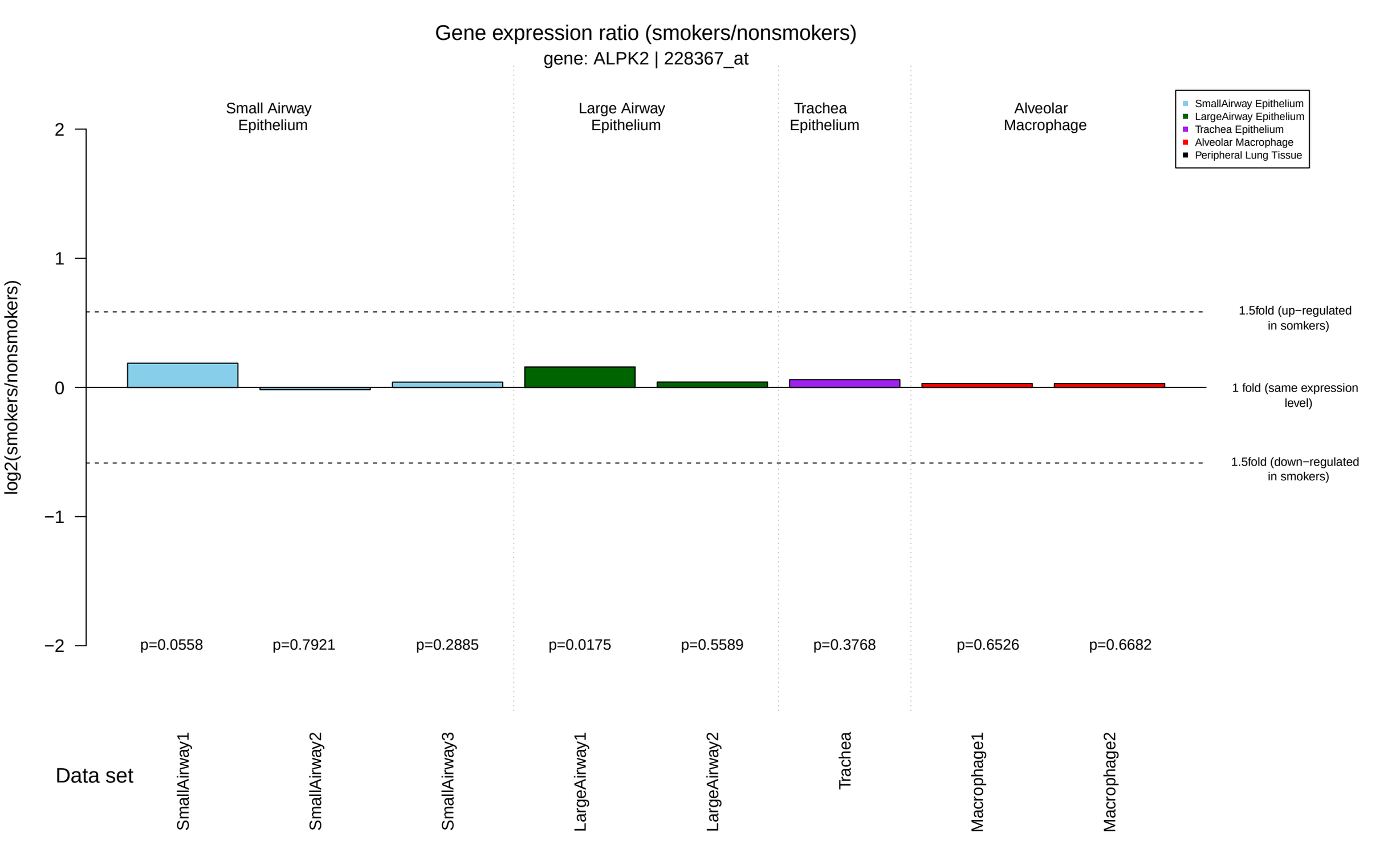
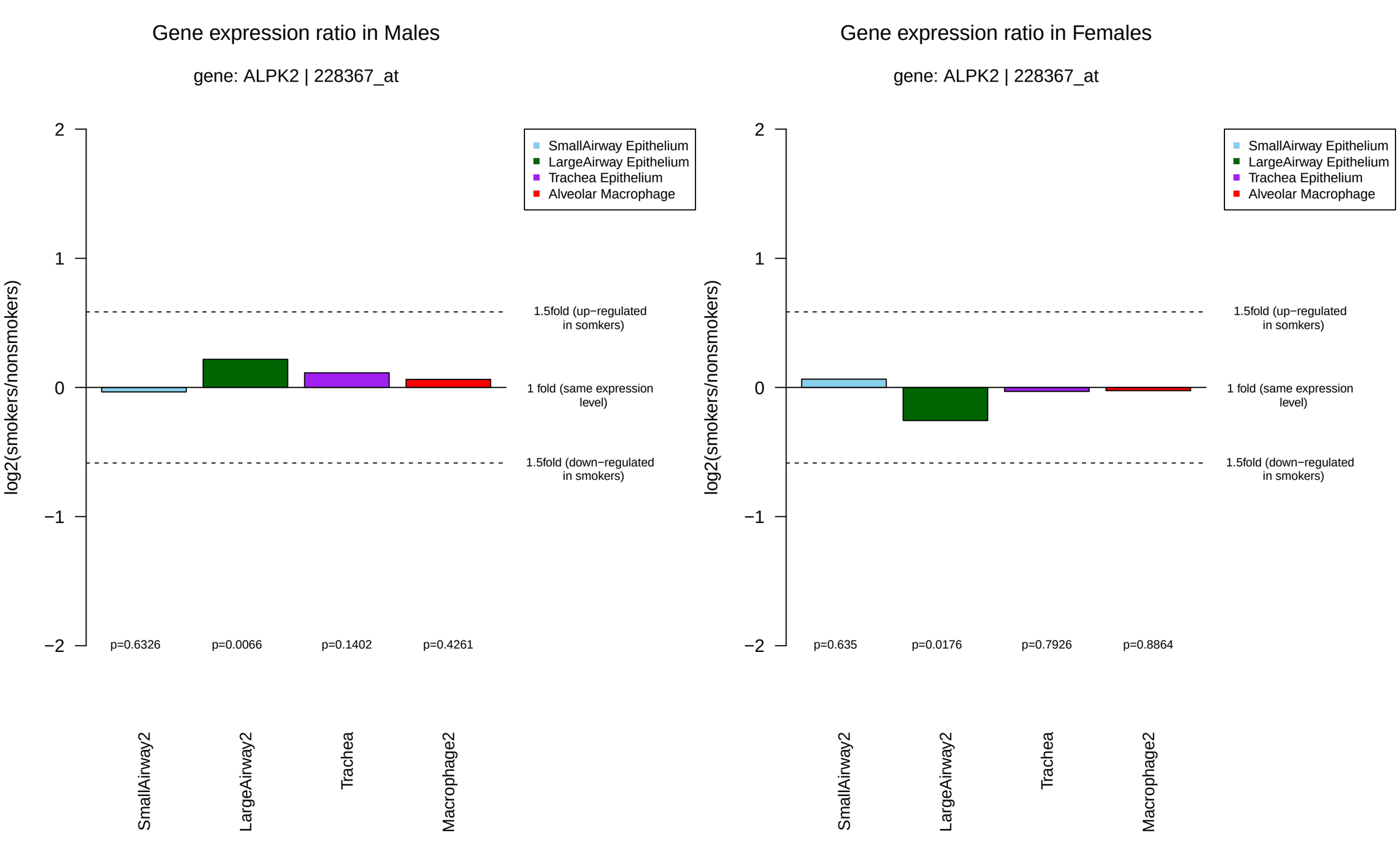
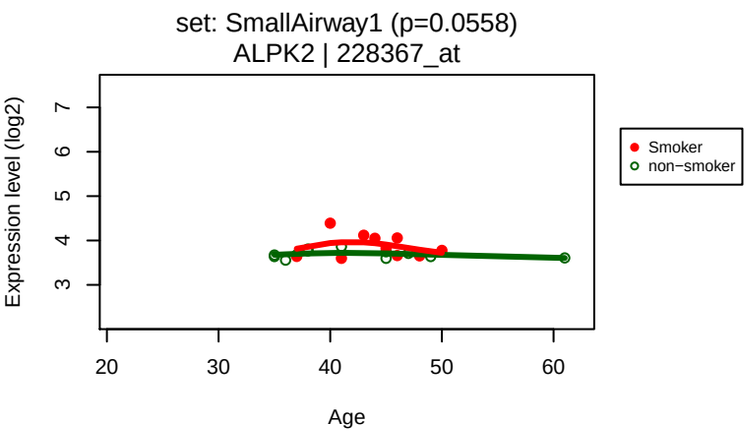
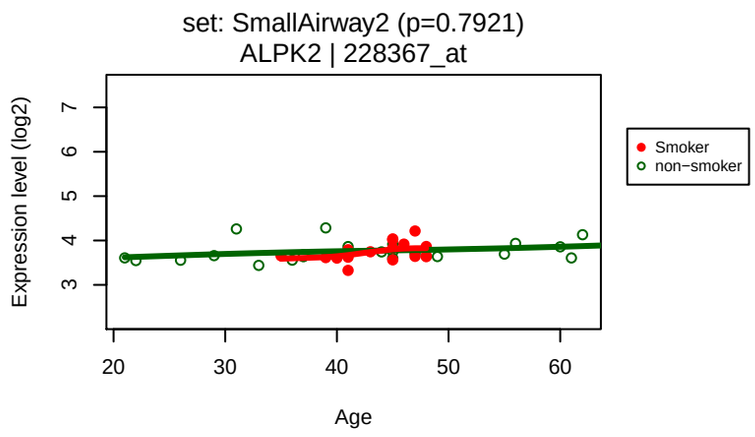
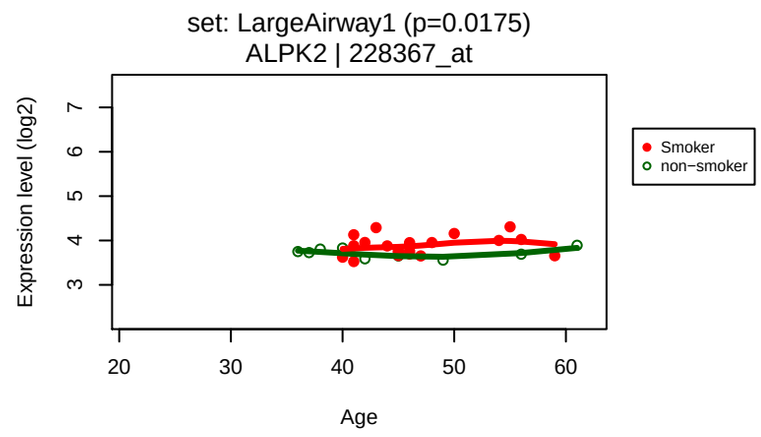

Gene expression of smokers and nonsmokers
gene: ALPK2 | 228367_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0558
p=0.7921
p=0.2885
p=0.0175
p=0.5589
p=0.3768
p=0.6526
p=0.6682
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

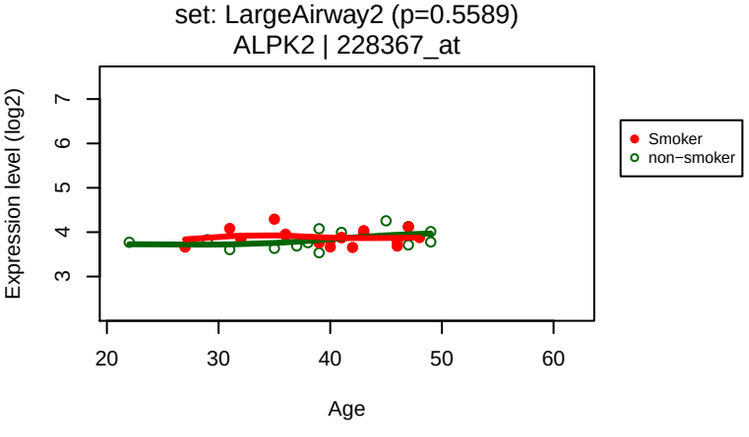

Gene expression ratio (smokers/nonsmokers)
gene: ALPK2 | 228367_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0558
p=0.7921
p=0.2885
p=0.0175
p=0.5589
p=0.3768
p=0.6526
p=0.6682
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

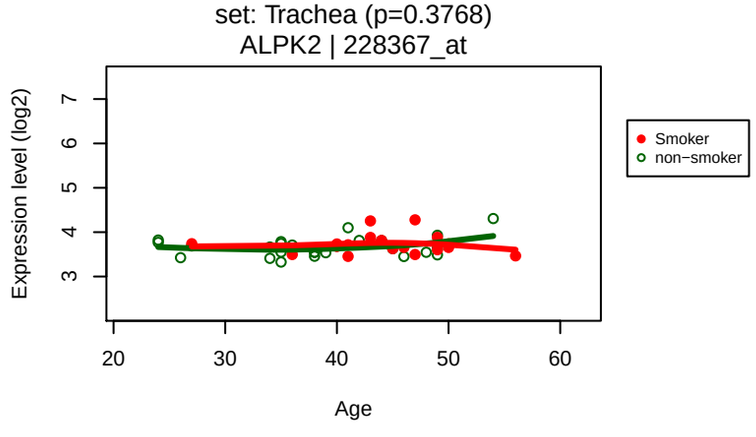

Gene expression ratio in Males
gene: ALPK2 | 228367_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6326
p=0.0066
p=0.1402
p=0.4261
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALPK2 | 228367_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.635
p=0.0176
p=0.7926
p=0.8864
SmallAirway2
LargeAirway2
Trachea
Macrophage2

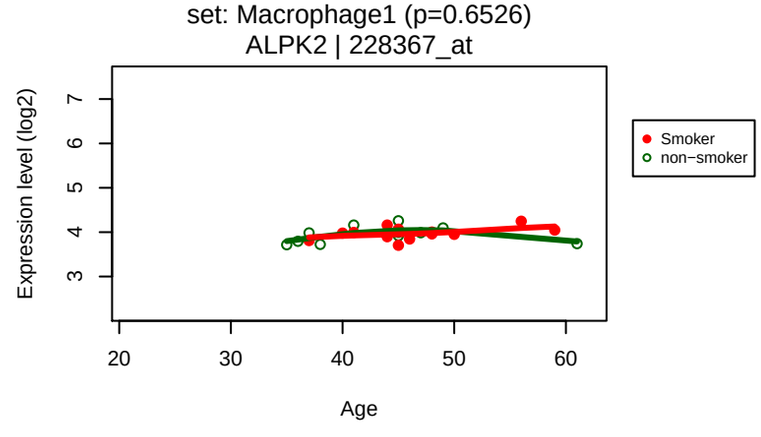

set: SmallAirway1 (p=0.0558)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker

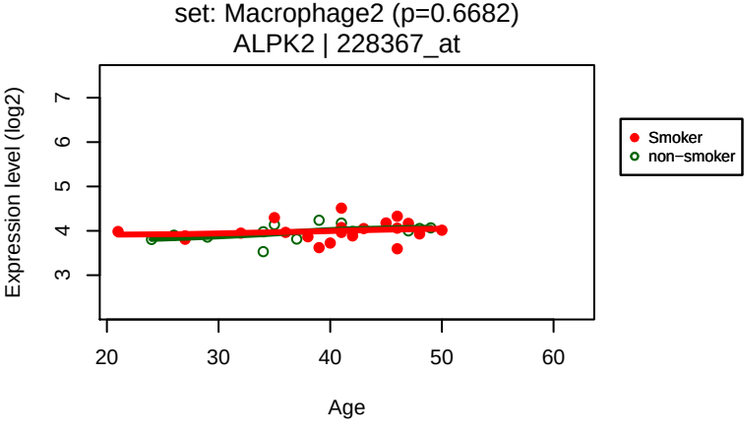

set: SmallAirway2 (p=0.7921)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker

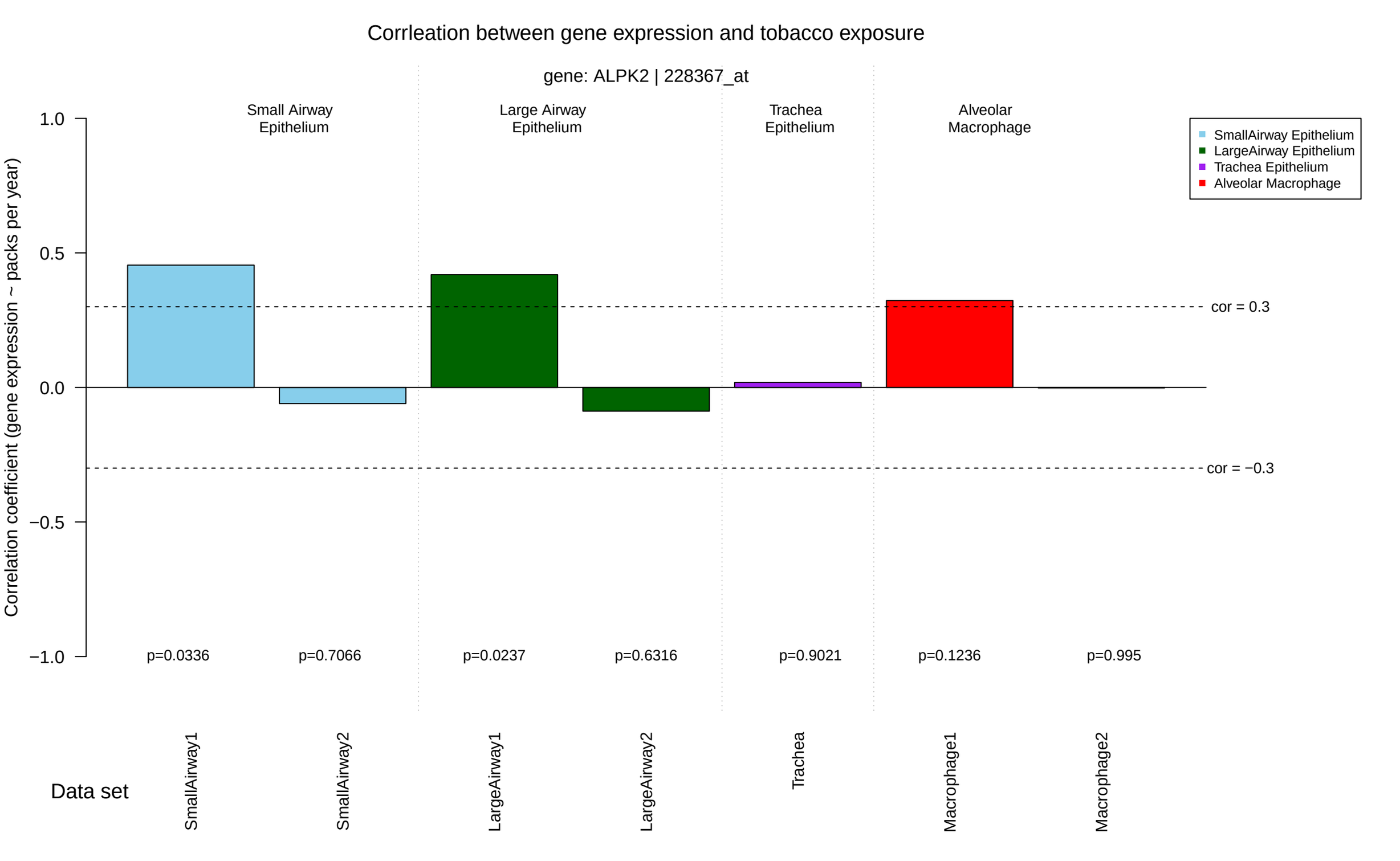

set: LargeAirway1 (p=0.0175)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5589)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3768)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6526)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6682)
ALPK2 | 228367_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALPK2 | 228367_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0336
p=0.7066
p=0.0237
p=0.6316
p=0.9021
p=0.1236
p=0.995
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2